Time–connectivity superposition and the gel/glass duality of weak colloidal gels

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Colloidal gels result from the aggregation of Brownian particles suspended in a solvent. Gelation is induced by attractive interactions between individual particles that drive the formation of clusters, which in turn aggregate to form a space-spanning structure. We study this process in aluminosilicate colloidal gels through time-resolved structural and mechanical spectroscopy. Using the time–connectivity superposition principle a series of rapidly acquired linear viscoelastic spectra, measured throughout the gelation process by applying an exponential chirp protocol, are rescaled onto a universal master curve that spans over eight orders of magnitude in reduced frequency. This analysis reveals that the underlying relaxation time spectrum of the colloidal gel is symmetric in time with power-law tails characterized by a single exponent that is set at the gel point. The microstructural mechanical network has a dual character; at short length scales and fast times it appears glassy, whereas at longer times and larger scales it is gel-like. These results can be captured by a simple three-parameter constitutive model and demonstrate that the microstructure of a mature colloidal gel bears the residual skeleton of the original sample-spanning network that is created at the gel point. Our conclusions are confirmed by applying the same technique to another well-known colloidal gel system composed of attractive silica nanoparticles. The results illustrate the power of the time–connectivity superposition principle for this class of soft glassy materials and provide a compact description for the dichotomous viscoelastic nature of weak colloidal gels.

weak colloidal gels | time–connectivity superposition | aluminosilicate | gels and glasses | relaxation time spectrum

Gelation is a dynamic process through which a weakly viscoelastic fluid suspension transforms into a solid-like material with a finite equilibrium elastic modulus (1, 2). This ubiquitous process plays an important role in synthetic biomaterials and polymer and inorganic hydrogels as well as colloidal suspensions such as paints, inks, and cosmetic products (3). In all these systems, gelation results in the formation of a space-spanning network whose strength depends on the chemical or physical nature of the underlying bonds.

In the case of physical gels, the interactions between the individual charge-stabilized colloids are well described by DLVO (Derjaguin, Landau, Verwey, and Overbeek) theory (4), which models the interactions as the sum of a weak attractive (van der Waals) and a repulsive (double layer) potential. Colloidal particle networks are often categorized as either glasses or gels (4–6) based on the particle volume fraction and the depth of the interaction potential energy (7). Depending on the relative contribution of each term, two limiting regimes of aggregation emerge, commonly referred to as diffusion-limited cluster aggregation (DLCA) and reaction-limited cluster aggregation (RLCA). Each scenario yields fractal gels with distinct values of mass fractal dimension d_f (8).

During the process of gelation, microstructural changes within the material strongly influence its rheological behavior. Accurate dynamic mechanical characterization of the time-evolving material properties has been achieved in chemical gels by arresting the polymerization process at different extents of reactions, corresponding to different states during the gelation process (3, 9, 10). Using the principle of time–cure or time–connectivity superposition (11), researchers have been able to develop universal master curves for the viscoelastic properties of these chemical gels over a wide range of frequencies (11–14), similar to those obtained using time–temperature superposition for nonreacting systems. The concept of time–connectivity superposition (11) relies on the idea that the fractal characteristics of the network are self-similar near the critical point and a change in the extent of reaction/gelation essentially results in a change of scale (9). A similar time–volume fraction superposition has been applied to strongly attractive colloidal suspensions by tuning the volume fraction and the attractive energy of solid particles (15). The validity of this principle has been successfully demonstrated also in microrheological measurements during gelation of similar biopolymer systems (16). Direct

Significance

Quantifying the viscoelastic properties of weak colloidal gels remains challenging due to the continuously aging nature of these pasty materials. Through time-resolved mechanical and structural spectroscopy and application of the time–connectivity superposition principle we construct the viscoelastic moduli of these gels over eight decades of effective material time. This protocol reveals the dual nature of the relaxation time spectrum with gel- and glass-like characteristics at large and small scales, respectively. The structure of the network at the critical gel point sets the underlying shape of the relaxation spectrum, which the aging material retains as a fixed skeleton. The observed nonergodic behavior suggests that the local arrest of clusters initiates from small scales and continuously evolves toward larger ones.

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measurements of the viscoelastic properties of weak colloidal gels during gelation are rare, mainly due to the very rapid rate of mutation and low stiffness at early stages of the gelation process.

Weak colloidal gels exhibit pronounced rheological aging broadly consistent with the predictions of the soft glass rheology (SGR) model (17, 18). A distinctive rheological signature of this aging behavior at long material timescales is a viscoelastic loss modulus $G''(\omega, t_{\text{age}})$ that continuously decreases with both the age of the system and with frequency in a power-law manner. Despite their widespread applications, little is known about the evolution of these materials at shorter material times, e.g., during the gelation process itself, including the validity of the time–connectivity superposition principle. Thus, an overall picture of the time-evolving viscoelastic properties of weak colloidal gels is still missing. Here we study the gelation of aluminosilicate gels and investigate the evolution in both their microstructural characteristics and their mechanical response at meso and macro scales through time-resolved small-angle X-ray scattering (TR-SAXS) and time-resolved mechanical spectroscopy (TR-MS), respectively. The evolution of aluminosilicate gel systems is of great interest in many applications including the production of zeolites and alkali-activated binders, also known as geopolymers (19, 20). The gelation process is initiated by mixing alkali silicate and alkali aluminate solutions together. Polycondensation reactions between silicate oligomers and aluminate monomers are initiated instantly and nanoscale aluminosilicate clusters grow rapidly with time to form a weak particulate gel (21). To address the short mutation time characterizing this gelation process, we use fast MS techniques (22), which allow us to measure the full linear viscoelastic response at different time points during and post gelation and correlate this information to the evolution of microstructural order. Our results and the associated self-similar constitutive framework provide insight to the gelation process for a wide class of geopolymers and the formation process of zeolites which pass through an amorphous stage before crystallization (23–26).

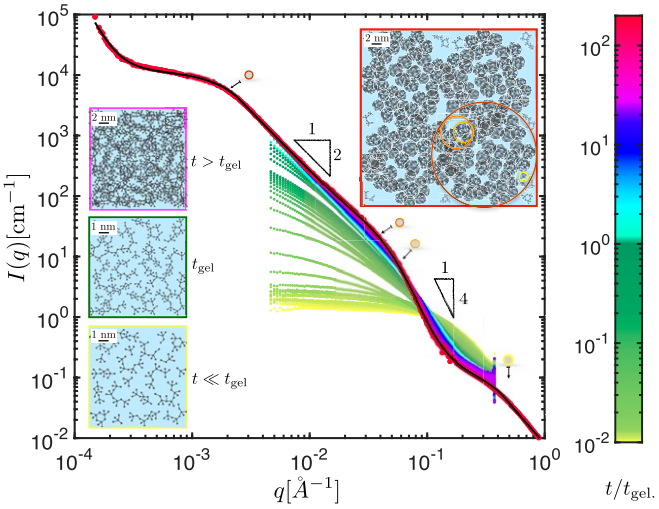


Fig. 1. Evolution of TR-SAXS intensity data during the gelation of a stoichiometrically unbalanced aluminosilicate gel. Here the sodium content is unbalanced in the pregel silicate and aluminate solutions and consequently the chemical reactions are of condensation type (see *SI Appendix* for more details). Intensity curves are colored from yellow to red in the color map displayed on the right based on their corresponding measurement time normalized by the gelation time (here t_{gel} = 3,200 s). (*Insets*) Qualitative sketches of the microstructure at various times after mixing. Polycondensation reactions are very fast prior to the gel point. We represent the important length scales in the qualitative sketch of the mature gel (*Top Right Inset* with red border) and map them on the measured spectra: The full USAXS/SAXS spectrum allows us to obtain quantitative information about the size of the clusters (dark orange circle $q \sim 0.0015\text{Å}^{-1}$), the correlation length (middle orange circle $q \sim 0.04\text{Å}^{-1}$), the size of the individual building blocks (light orange $q \sim 0.06\text{Å}^{-1}$) and the size of the remaining dissolved silicate entities into the solution (yellow circle $q \sim 0.5\text{Å}^{-1}$). A mass fractal dimension of $d_f = 2$ can be determined for the clusters.

Results

Mesoscale Characterization by TR-SAXS. The structural kinetics of gelation over a wide range of length scales are shown in Fig. 1 for a stoichiometrically unbalanced aluminosilicate gel (see *SI Appendix* for the corresponding case of a stoichiometrically balanced gel, which yields similar results). TR-SAXS experiments are initiated shortly after the silicate and aluminate solutions are mixed together in a T-mixing channel (*Materials and Methods* and *SI Appendix*, Fig. S2). The condensed silicate and aluminate monomers react instantaneously to form aluminosilicate building blocks, which are weakly attractive. The first spectrum, measured less than 10 s after mixing, shows a Guinier plateau with a radius of gyration around 8 Å, which is bigger than the radius of gyration for a dissolved silicate molecule (*SI Appendix*, Fig. S3). This indicates that even at this early stage ($t \ll t_{\text{gel}}$) discrete aluminosilicate building blocks are formed (yellow inset sketch in Fig. 1). These very fast polycondensation reactions continue and the aluminosilicate building blocks grow with high polydispersity, as indicated by the broad evolution of the scattered intensity in the lower q-range ($q \leq 0.05\text{Å}^{-1}$). Concomitantly, the small dissolved silicate species are consumed as evidenced by the decreasing scattering intensity in the higher q-range for $q > 0.1$ Å^{-1} (*SI Appendix*, Fig. S6). These colloidal aluminosilicate entities aggregate with each other, eventually forming a percolating network with a fractal dimension close to $d_f \simeq 2$ at a gelation time $t_{\text{gel}} \simeq 3{,}200$ s (as indicated by the green inset sketch in Fig. 1). After the gel point, a local correlation peak appears at larger length scales ($q = 0.04\text{Å}^{-1}$) and its intensity increases with time (see the corresponding Kratky plot in *SI Appendix*, Fig. S6), indicating that local liquid ordering develops. To enlarge the microstructual observation window, ultra-small-angle X-ray scattering (USAXS) measurements were also performed on the mature aluminosilicate gel 2 d after synthesis and are shown by the red curve in Fig. 1. At very small scales, corresponding to $q \geq 0.2\text{Å}^{-1}$, the spectra are dominated by scattering of small silicate entities ($R_{g,0} = 4$Å) that have not participated in the polycondensation reaction with aluminates and are still dissolved in the solvent. The power-law behavior $I \sim q^{-4}$ for $0.05 \leq q \leq 0.2\text{Å}^{-1}$ indicates a Porod regime with smooth interface between the aluminosilicate building blocks, ($R_{g,1} = 11.5$Å) and the solution. At larger scales, the network formed by the aggregation of these individual building blocks gives rise to fractal clusters ($R_{g,2} = 680$ Å), as reflected in the broad power-law behavior for $10^{-3} \leq q \leq 10^{-1}$ Å^{-1}. The power-law slope in this range indicates a mass fractal dimension of $d_f \simeq 2$ which is typical of RLCA processes (8). At even larger scales, these fractal structures form bigger agglomerates and the scattering from the interfaces between these large agglomerates is visible down to the lowest values of accessible wave vectors; for $q \leq 3 \times 10^{-4}$ Å^{-1}, the scattering intensity again shows a Porod regime with a rather steep power-law exponent (close to q^{-4}), indicating smooth interfaces of these large agglomerates. The USAXS/SAXS scattering data are well described by a combination of Guinier–Porod models, whose fitting parameters allow us to estimate the volume fraction of particles participating in the gel network to be $\phi \approx 1.5\%$, consistent with the theoretical prediction based on the initial concentration of aluminum (see *SI Appendix* and *SI Appendix*, Table S1 for details). The

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same analysis was carried out for the stoichiometrically balanced aluminosilicate gel and we observe similar trends (*SI Appendix*, Figs. S5 and S7), illustrating that, despite slight differences in the chemical balance of the initial reactions, these weak colloidal gels are fractal clusters that are formed through reaction-limited cluster aggregation of smaller building blocks. Recent structural studies on weak colloidal gels with short-range attraction have also reported similarly dense fractal structures with $d_f \simeq 2$ (20, 27).

Rheology by TR-MS. To connect the structural and rheological evolution of these aluminosilicate gels, we also performed time-resolved MS experiments during the gelation process. If we first consider an unbalanced gel at a fixed mechanical probe frequency ω_0, measurement of the temporal evolution of the storage and loss moduli (Fig. 2*A*) shows that the pregel is initially a viscoelastic liquid with $G'' > G'$. As the gel point is approached, both viscoelastic moduli increase rapidly: The elastic modulus grows more than a hundredfold and the characteristic mutation timescale can be estimated to be $\tau_{\text{mut}} = \{\max[d \ln G'/dt]\}^{-1} \approx 3 - 15$ s. Beyond the gelation event, the elastic modulus continues to grow toward a plateau, while the loss modulus passes through a local maximum before gradually decreasing. In order to measure the instantaneous mechanical relaxation spectrum of this aluminosilicate gel during the entire process of gelation, we need to use probe signals that can be completed in a timescale as short as τ_{mut}. We achieve this by application of a sequence of optimally windowed chirp (OWCh) signals (22) (see *Materials and Methods* and *SI Appendix* for more details) with an adaptive duration that guarantees rapid acquisition timescales even close to the gel point. With these multifrequency chirp signals we are able to follow the temporal evolution of the viscoelastic properties of the pregel, critical gel, and postgel states over a range of frequencies. For each measured frequency spectrum (at a time t_i) we also compute the ratio of loss to elastic modulus, given by $\tan\delta = G''(\omega, t_i)/G'(\omega, t_i)$, where $\delta(\omega, t_i)$ is the phase angle, and follow its progressive evolution with the age of the system. These data allow us to unambiguously identify the true gel point, defined as the time $t_i = t_{\text{gel}}$ when $\tan\delta$ becomes independent of frequency (green line in Fig. 2*C*). At this critical time both moduli show a power-law dependence with ω (i.e., $G', G'' \sim \omega^{\alpha}$), due to the self-similar and scale-free nature of the percolating network formed at the gel point (10). For the unbalanced aluminosilicate gel we find that $\alpha = 0.48 \pm 0.02$. It has been shown that this behavior of the critical gel can be well described with a rheological model based on a single fractional element, known as a spring-pot (28–30). However, beyond the critical gel point, $\tan\delta(\omega, t)$ does not remain constant in aluminosilicate gels. The phase angle decreases with time and gradually shows a distinct frequency-dependent behavior (Fig. 2*C*). For all of the weakly attractive colloidal gels studied, we observe that far beyond the gel point (i.e., in the mature gel) $\tan\delta$ asymptotically approaches a power-law relationship $\tan\delta \sim \omega^{-\alpha}$, where α is the exponent determined at the gel point. Such behavior is in stark contrast to that observed in chemically cross-linked gels, where postgel values of $\tan\delta$ increase with frequency (10). In this mature state ($t \gg t_{\text{gel}}$), the viscoelastic spectrum of the colloidal gel displays a weakly increasing storage modulus $G'(\omega, t_{\text{age}})$ and a power-law decrease in $G''(\omega, t_{\text{age}})$ with both sample age and frequency (Fig. 2*B*). This behavior indicates lower values of viscoelastic dissipation at smaller length and timescales (corresponding to higher frequencies). As shown in *SI Appendix*, Fig. S8, and also Fig. 3*A* for the stoichiometrically balanced formulation, the dissipation progressively decreases with the age of the gel and the characteristic timescale of the network τ_c increases with time in a nonergodic manner; such that the system is not time-transitionally invariant (31). Theoretical frameworks such as the SGR model can capture this viscoelastic behavior at long aging times through a slowing down in the hopping dynamics of particles in an evolving energy landscape (17, 18). This limit of weak ergodicity breaking in the SGR model does not, however, provide predictions of the rheological response very

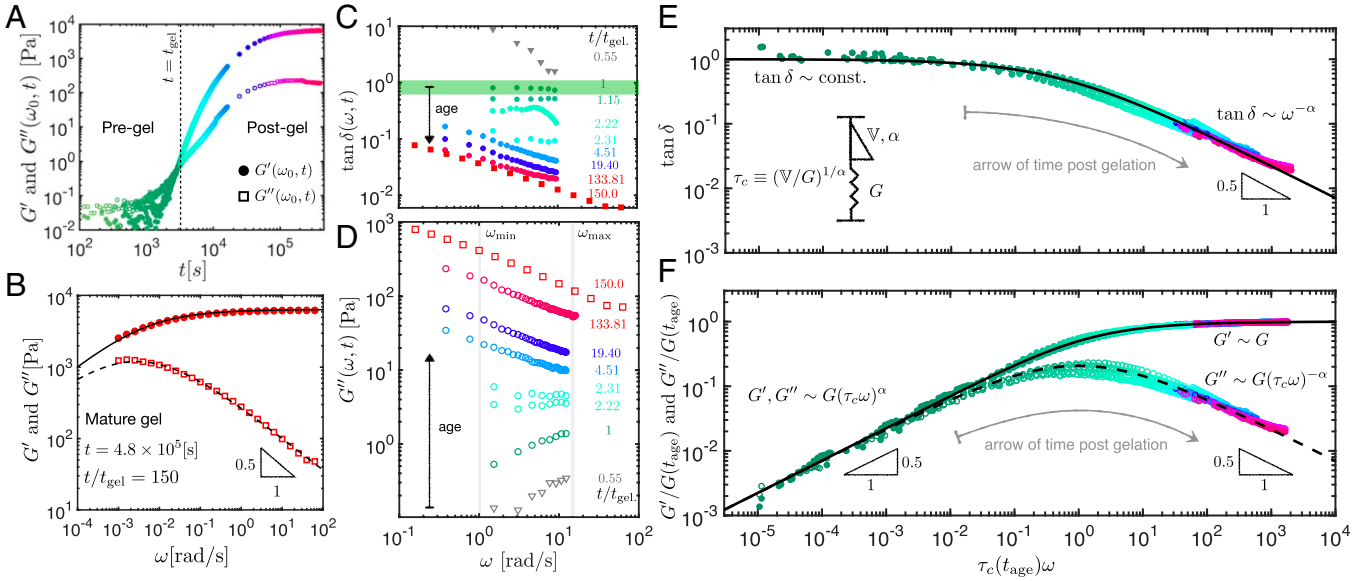


Fig. 2. TR-MS measurements on an aluminosilicate gel (with unbalanced stoichiometry). We use the same color code as in Fig. 1. (*A*) Linear viscoelastic moduli G' and G'' vs. time at a fixed probe frequency $\omega_0 = 10$ rad/s. (*B*) Viscoelastic spectrum of the mature gel determined at $t_{\text{age}} = 4.8 \times 10^5$s, $t/t_{\text{gel}} = 150$: G' (•) and G'' (○). The solid and dashed lines correspond to the predictions of the fractional Maxwell gel (FMG) model with $\mathbb{V} = 4.62 \times 10^4$ Pa·s$^{\alpha}$, $\alpha = 0.5$ and $G = 6.64 \times 10^3$ Pa (Eq. **2**). (*C* and *D*) Plots of measured values of $\tan\delta = G''/G'$ and loss modulus G'' vs. frequency at various sample ages during gelation. The gel point, characterized by a frequency-independent phase angle, is highlighted inside the green band in *C*. (*E* and *F*) Evolution in phase angle and scaled viscoelastic moduli vs. normalized frequency $\omega\tau_c$. The "arrow of time post gelation" indicates the evolution of the aluminosilicate gel toward its mature state. Asymptotic behavior of the viscoelastic material functions at low and high frequency is indicated on the graph (*F*). The black lines correspond to the predictions of the FMG model (Eq. **2** with $\alpha = 0.5$).

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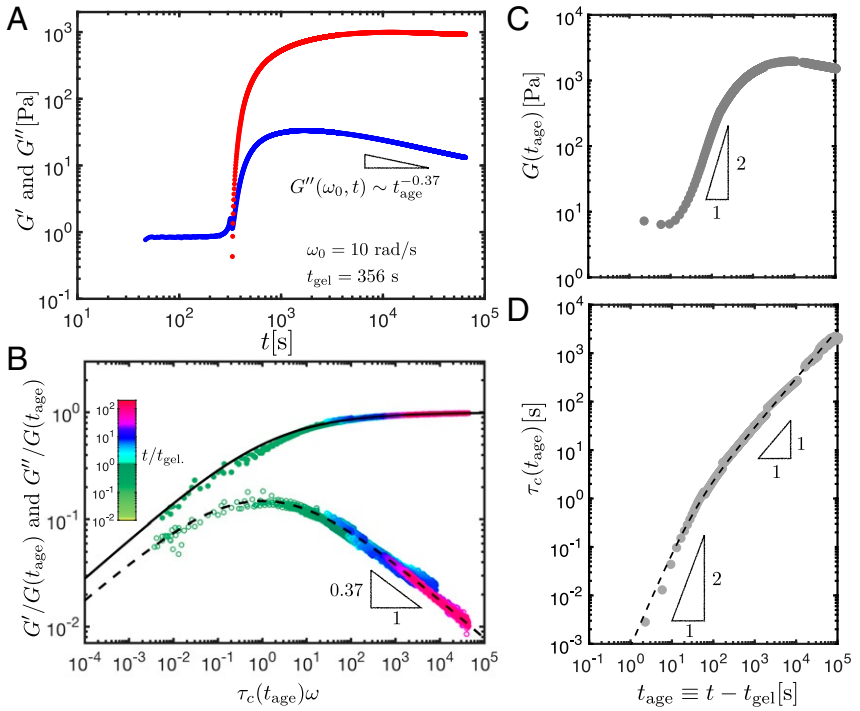


Fig. 3. TR rheology measurements on the stoichiometrically balanced aluminosilicate gel. For this composition, gelation happens even more rapidly and $t_{\rm gel} = 356$ s. (*A*) Temporal evolution of storage and loss modulus at $\omega_0 = 10$ rad/s. At long times the loss modulus of the gel shows an aging behavior with a power-law decay $G''(\omega_0, t) \sim t^{-0.37}$. (*B*) Master curve for the viscoelastic moduli. Filled and open symbols show values of the storage and loss modulus, respectively; solid and dashed black lines represent the predictions of the three-parameter FMG model with exponent $\alpha = 0.37$. (*C* and *D*) Evolution of the vertical and lateral shift factors, respectively, given by the characteristic elastic modulus scale $G(t_{\rm age})$ and relaxation time $\tau_c(t_{\rm age})$ for this gel.

close to the gel point, nor the behavior at very low frequencies. Other multiparameter models such as the BSW (Baumgaertel–Schausberger–Winter) model (32), inspired from glass transition studies (33), and models for pasty materials (34) face similar challenges in describing the full rheological response over a wide range of age time and frequencies. Thus, we adopt a canonical mechanical framework that accurately captures the physical essence of the aforementioned models in terms of both the scale-free power-law nature of the underlying percolating network and the progressively lower levels of viscoelastic dissipation at smaller time scales. We also introduce aging and weak ergodicity breaking into our model by introducing a time-evolving effective material timescale $\tau_c(t_{\rm age})$ (31, 35) to characterize the elasticity of the network. The resulting mechanical model comprises of a Hookean spring (with shear modulus G) in series with a fractional (or spring-pot) element characterized by two parameters: an exponent α and a quasi-property $\mathbb{V}$ that characterize the complete viscoelastic frequency response of the fractal network (29). This three-parameter model, analogous to the modified Cole–Cole model reported in refs. 36 and 37, is a reduced version of the more general fractional Maxwell model (28) with a purely elastic gel-like response at high frequencies, which we thus refer to as the fractional Maxwell gel (FMG) model. As we show later, the constitutive equation of the FMG model can be written in terms of a Boltzmann integral over a continuous spectrum of relaxation times. However, it can also be written compactly as a fractional differential equation:

$$\sigma + \frac{\mathbb{V}}{G}\frac{d^\alpha \sigma}{dt^\alpha} = \mathbb{V}\frac{d^\alpha \gamma}{dt^\alpha}, \qquad [1]$$

which upon Fourier transformation yields the following expression for the complex viscoelastic modulus:

$$G^*(\omega) = G'(\omega) + iG''(\omega) = G(t_{\rm age})\frac{(i\omega\tau_c)^\alpha}{(i\omega\tau_c)^\alpha + 1}, \qquad [2]$$

where $G(t_{\rm age})$ is the (time-evolving) elastic modulus of the gel and $\tau_c(t_{\rm age}) = (\mathbb{V}/G)^{1/\alpha}$ is a characteristic timescale that divides the viscoelastic response of the aging material into two regions. On long timescales $\tau > \tau_c(t_{\rm age})$ (or equivalently frequency scales $\omega\tau_c(t_{\rm age}) < 1$) it predicts an age-dependent power-law behavior similar to a critical gel with a time-evolving quasi-property $\mathbb{V}(t_{\rm age}) = G\tau_c^\alpha$, whereas at short timescales $\tau < \tau_c(t_{\rm age})$ (or $\omega\tau_c(t_{\rm age}) > 1$) it predicts a frequency-independent elastic modulus and a power-law decrease in the loss modulus ($G' \simeq G$ and $G'' \simeq (G^2/\mathbb{V})\omega^{-\alpha} = G/[\omega\tau_c(t_{\rm age})]^\alpha$).

Intriguingly, as we show in Fig. 2*B*, the viscoelastic moduli of the mature aluminosilicate gel (measured at aging timescales of $t_{\rm age} \simeq O(10^5)$ s) are also well described by the theoretical predictions of the FMG model with the same power-law exponent $\alpha = 0.5$, determined independently and tens of thousands of seconds earlier from measurements at the critical gelation time. However, the very large value of the characteristic timescale for the mature gel (here $\tau_c \simeq 10^3$ s) means that the viscoelastic properties are only measured at frequencies above $1/\tau_c$ and the terminal response of the mature gel cannot easily be observed using small-amplitude oscillatory shear tests. For example, measurements at frequencies $\omega \leq 10^{-4}$ rad/s require timescales of $O(2\pi/\omega) \sim 10^5$ s, which approach or exceed the age of the gel. This difficulty of fully characterizing aging materials was first considered by Struik (31) in the context of slowly aging glassy polymers. Due to the continuous aging of our soft glassy material we cannot see sufficiently far back in time to

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understand the gelation process if we only probe the gel in its mature state.

To study the material closer to this finite time horizon, we use the ideas of an effective material time (31) and time–connectivity superposition (11–13) to follow the evolution in the mechanical properties continuously through the gel point and beyond. As shown by our SAXS studies, the characteristic length scales and their relative scattering intensities vary systematically with time throughout the gelation process. We may expect (12, 13) this structural progression to be reflected through evolution in the full age-dependent relaxation spectrum of the gelling material $H(\tau; t_{age})$, which is compactly captured in the characteristic relaxation time $\tau_c(t_{age})$ and modulus $G(t_{age})$ of the gel. To deconvolve the evolution of these two material functions that parameterize the underlying relaxation spectrum, we employ a method that is broadly utilized in time–temperature superposition studies and focus first on the evolution of the phase angle, $\tan\delta = G''/G'$. This parameter is independent of the gel strength and hence allows us to focus exclusively on the evolution of the characteristic timescales of the evolving network. The self-similar evolution in the relaxation spectrum is revealed by the observation that all of the measured $\tan\delta$ curves can be collapsed onto a master curve by applying a simple age-dependent horizontal shift to the viscoelastic properties measured using exponential chirps during gelation. As shown in Fig. 2*E*, this horizontal shifting results in a master curve for $\tan\delta(t_{age})$ with a rescaled dimensionless frequency $\omega\tau_c(t_{age})$ on the abscissa. The shift factor $a_t \equiv \tau_c(t_{age})$, calculated for each spectrum, captures the evolution in the characteristic timescale of the aging network. This master curve describes the entire gelation process for the unbalanced aluminosilicate gel as the material evolves from a critical gel to its mature state (indicated by the arrow of time in Fig. 2*E*). We observe an asymptotic limit of $\tan\delta = \tan(\alpha\pi/2) = 1$ for $\omega\tau_c \ll 1$, capturing both the frequency-independent value of $\tan\delta$ at the critical gel point and an exponent value of $\alpha = 0.5$, as well as a limit of $\tan\delta \sim (\omega\tau_c)^{-\alpha}$ for $\omega\tau_c \gg 1$, which agrees very well with the rheological behavior of the mature gel (cf. Fig. 2*B*) and the prediction of our three-parameter model.

Having determined the evolution of the timescale for the microstructural network, the strength of the aging gel can be quantified from the evolution of the viscoelastic moduli as a function of the dimensionless reduced frequency $\omega\tau_c$. As shown in Fig. 2*F*, we construct a master curve for the complex moduli by shifting each instantaneous mechanical spectrum vertically with a shift factor $b_t \equiv 1/G(t_{age})$, where $G(t_{age})$ represents the instantaneous stiffness scale of the aging network. These master curves clearly show that the time–connectivity superposition principle applies here: Once the gel structure is formed at the critical gel point, the evolving network retains the fractal geometry of the original structure and evolves with time in a self-similar manner. The master curves represent the viscoelastic behavior of the material over a very wide range of timescales that span eight orders of magnitude, a range that is experimentally inaccessible in conventional tests with mature gels. The same superposition principles also apply to stoichiometrically balanced aluminosilicate gels, as we show in Fig. 3 *A* and *B*. The change in chemistry is reflected solely by a change in the exponent α characterizing the critical gel. For this system $\alpha = 0.37$ but the three-parameter FMG framework again captures the evolution in the full mechanical fingerprint of the network structure at all times from gelation onward.

The observation that the postgelation mechanical spectra retain the same shape at all points in time, which is controlled by a single parameter of the FMG framework (the exponent α) also significantly simplifies our analysis for this class of mutating colloidal gels. The entire temporal evolution of the aging system can be characterized through time-dependent changes of just two physical properties of the underlying network: the instantaneous characteristic timescale $\tau_c(t_{age})$ and the characteristic stiffness $G(t_{age})$ of the underlying relaxation spectrum. These parameters are embodied respectively in the horizontal and vertical shift factors (cf. Fig. 3 *C* and *D* and *SI Appendix*, Fig. S8).

Discussion

Structural and Dynamical Properties of the Gel. To connect the structural and dynamical properties of the gels, we use the relationship introduced by Muthukumar (38) that links the power-law exponent α with the fractal dimension of the network at the gelation point. This model, originally developed for branched polymer systems and later used for other gelling systems including colloidal gels, predicts that, in the limit of dominant hydrodynamic screening interactions between the constituents, for a three-dimensional gel $\alpha = 3(5-2d_f)/2(5-d_f)$, where d_f is the mass fractal dimension of the network. Using this equation, we can determine d_f for our systems based on the measured values of the exponent α and find that $d_f = 2.1 \pm 0.05$ for the balanced gel ($\alpha = 0.37$), and $d_f = 2.0 \pm 0.05$ for the unbalanced gel ($\alpha = 0.5$). These results are fully consistent with the fractal dimensions obtained from scattering experiments (Fig. 1 and *SI Appendix*, Fig. S7). More generally, using Muthukumar's relation for $0 \le \alpha \le 0.5$ yields a range of possible fractal dimensions $2.0 \le d_f \le 2.5$ consistent with the RLCA mechanism that is characteristic of these weakly attractive gels. Another recent investigation has also concluded that aluminosilicate gels are formed through RLCA for a range of compositions (20). This analysis suggests that our results should hold for all gels that are in the so-called solid-cluster limit, i.e., gels that display a long-time elastic-like behavior ($G' \ge G''$ at low frequencies) with $\alpha \le 0.5$ (5).

Aging. Both balanced and unbalanced gels display a nonergodic response characterized by the slow decrease of G'' at long time (Figs. 2*A* and 3*A*). The effect is more visible for the balanced gel, in which gelation occurs earlier. For this system we can clearly see that the loss modulus at long times ($t_{age} \sim 10^4$ s) decreases as a power law of time, i.e., $G''(\omega_0, t) \sim t_{age}^{-0.37}$, with an exponent that is identical to the exponent α of the critical gel observed at $t_{gel} = 356$ s. The identical values of the two power-law exponents determined from aging of the mature gel and the linear viscoelastic response of the material at the gel point can be understood within the self-similar framework of the FMG model. As shown in Fig. 3*D*, at early times after the gelation ($t_{age} \le 10^2$ s) the characteristic timescale τ_c of the network obtained from time–connectivity superposition increases very rapidly with the gel's age, first increasing quadratically [which is similar to what is referred as hyperaging in physical gels (35)]. At longer times, however ($t_{age} \ge 10^3$ s), the characteristic time increases linearly with the gel's age, i.e., $\tau_c \sim t_{age}$, and the elastic modulus approaches a plateau value G_∞ (cf. Fig. 3*A*). Thus, as the gel matures, the quasi-property $\mathbb{V}$ characterizing the collective time-dependent response of the self-similar relaxation modes in the network scales as $\mathbb{V}(t_{age}) \sim G_\infty t_{age}^{\alpha}$. Within our FMG framework, the imaginary component of Eq. **2** results in the loss modulus scaling as $G''(\omega) \sim (G^2/\mathbb{V})\omega^{-\alpha}$ for mature gels, and hence substituting $\tau_c = (\mathbb{V}/G)^{1/\alpha}$ we find $G''(\omega, t) \simeq G_\infty(\omega t_{age})^{-\alpha}$. This asymptotic consideration shows that the FMG framework consistently incorporates the temporal evolution of model parameters ($G(t_{age}), \tau_c(t_{age})$) and provides accurate predictions for the long-time aging response of these weak colloidal gels.

From a modeling perspective, the aging behavior shares some common features with the dynamics of mode-coupling theories and the SGR model developed for soft glassy materials that exhibit weak ergodicity breaking (18). In the SGR model,

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the relaxation of stress in the material is described in terms of an activated hopping process over a distribution of energy barriers. The dynamics of this process is governed by an effective noise temperature x, and a nonergodic pseudo-glass regime is predicted when $x < x_g$, where x_g is the average height of the energy barriers. In this limit, the SGR model predicts that G'' also decays as a power law with both increasing time and frequency, i.e., $G''(\omega, t) \sim (\omega t_{\text{age}})^{-m}$, where $m = 1 - x/x_g$ (18). The similarity between this prediction and our results for mature aluminosilicate gels suggests that aging arises from weak ergodicity breaking and suggests a glass-like regime at short timescales (large frequency scales). Comparing the aging exponents between the two models, we can see a direct relation between our frequency exponent α and the noise temperature ($x/x_g = 1 - \alpha \leq 1$), which reinforces the intimate connection between a critical gel and rheological aging in a soft glass that is embodied in the time–connectivity superposition principle.

From an experimental point of view, the viscoelastic response of aluminosilicate gels that we observe at high frequencies $\omega\tau_c(t_{\text{age}}) \gg 1$ is common to many other aging systems (39–41), which fall in the category of weakly attractive colloidal gels that also exhibit nonnegligible repulsion among constituents, also referred to as pasty-phase materials in some references (34). However, the low-frequency response of such pasty gels has so far been experimentally inaccessible due to material aging and the long acquisition times required for measurements. Here, we have shown through time–connectivity superposition and TR rheology that, on sufficiently long timescales, the terminal response of these materials is identical to the scale-free mechanical spectrum of a critical gel. Importantly, we note that this is completely contrary to the response observed for chemically cross-linked polymer gels and colloidal gels with strong interparticle attractions (11). For these systems it is the short-time (or high-frequency) response of the mature gel that remains gel-like, while a finite terminal elastic modulus G_e and glass-like response emerges at low frequencies.

The Relaxation Time Spectrum for Weak Colloidal Gels. To investigate in more detail the nature of these aluminosilicate gels, we consider the relaxation modulus of the network. For a time-dependent viscoelastic material this is the net sum of contributions to the stress in the Boltzmann integral resulting from modes with different relaxation timescales. In the continuous limit, this can be written in integral form as

$$G(t, t_{\text{age}}) = \int_0^\infty e^{-t/\tau} H(\tau, t_{\text{age}}) \frac{d\tau}{\tau}, \qquad [3]$$

where the relaxation time spectrum, $H(\tau, t_{\text{age}})$, represents the distribution of relaxation modes in the mechanical response of the viscoelastic material (37). Using a Stieltjes integral transformation, an analytical expression for the relaxation spectrum can be obtained from the measured viscoelastic moduli as $H(\tau) = (2/\pi)\text{Im}\left\{G'\left(\tau^{-1}\exp(i\pi/2)\right)\right\}$ (37). For our FMG model with an elastic modulus $G(t_{\text{age}})$, a characteristic timescale $\tau_c(t_{\text{age}})$, and a frequency exponent α the relaxation spectrum can be expressed in the self-similar form:

$$\frac{H(\tau, t_{\text{age}})}{G(t_{\text{age}})} = \frac{1}{\pi} \frac{\sin(\pi\alpha)}{x^{-\alpha} + 2\cos(\pi\alpha) + x^{\alpha}}, \qquad [4]$$

where $x = \tau/\tau_c(t_{\text{age}})$. Combining the results from Eqs. **2** and **4**, we can compute a master curve for the continuous relaxation spectrum of our time-evolving gels based on the viscoelastic spectra determined experimentally using exponential chirps during the gelation process. The results for the unbalanced and balanced aluminosilicate gels are shown in Fig. 4 *A* and *B*, respectively. Similar to the loss modulus $G''(\omega, t_{\text{age}})$, the relaxation spectra for these two gels are symmetric about a characteristic (age-dependent) value τ_c. For short relaxation

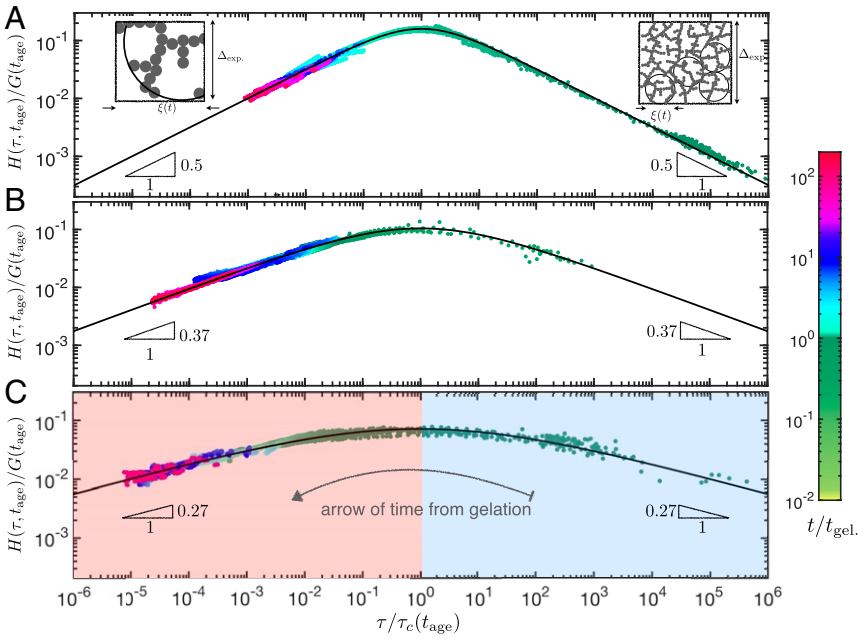


Fig. 4. Dimensionless master curves for the evolving relaxation time spectrum $H(\tau, t_{\text{age}})$ of (*A*) the unbalanced, (*B*) the balanced aluminosilicate gel, and (*C*) the Ludox system. Data points are extracted from time–connectivity superposition analysis of successive mechanical spectra measured using exponential chirps. Solid lines show the theoretical predictions (Eq. **4**). The gel's path from the critical gel point to a mature gel conforms with a progressive transition from the right side (where $\tau/\tau_c \gg 1$) to the left side (where $\tau/\tau_c \ll 1$) of the relaxation spectrum. The upper right inset is a cartoon representation of the case where the length scale of our experimental window is larger than the typical floc size $\Delta_{\text{exp.}} > \xi(t_{\text{age}})$ and consequently a gel-like behavior with a negative power-law exponent ($\tau^{-\alpha}$) is observed. The opposite scenario occurs in the upper left inset and consequently a glass-like behavior with a positive power-law exponent (τ^{α}) is observed.

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modes $\tau/\tau_c(t_{age}) \ll 1$ the spectrum increases as a power law with a positive exponent $H(\tau, t_{age}) \sim G(t_{age})\,[\tau/\tau_c(t_{age})]^{\alpha}$, and for long modes with $\tau/\tau_c(t_{age}) \gg 1$ it decays symmetrically with a negative exponent $H(\tau, t_{age}) \sim G(t_{age})\,[\tau/\tau_c(t_{age})]^{-\alpha}$.

To investigate the generality of this result, we also studied the gelation of a colloidal gel of silica nanoparticles (Ludox, radius 13 nm) (42). Similar to our aluminosilicate gels, silica nanoparticles are known to aggregate and form reaction limited clusters with fractal dimension $d_f = 2.1$ (41, 43). We again monitor the linear viscoelastic properties during gelation by TR rheology to construct a master curve for the corresponding relaxation spectrum (Fig. 4*C*). Once again, the relaxation time spectrum, observed postgelation (here $t_{gel} = 480$ s) shows the same shape observed for the aluminosilicate gels; however, for this system the critical gel is characterized by a weaker frequency dependence with $\tan\delta(t_{gel}) = 0.45$ and an exponent $\alpha = 0.27$.

The shape of the relaxation spectra reveals that all of these weak colloidal gels are thus intrinsically characterized by a dual viscoelastic nature. At long timescales ($\tau/\tau_c(t_{age}) \gg 1$), the intensity of the relaxation spectrum decays with the timescale of each relaxation mode as $H(\tau, t_{age}) \sim G(t_{age})\,[\tau/\tau_c(t_{age})]^{-\alpha}$, indicating a progressively smaller contribution from long-timescale modes (and larger structures) to the overall relaxation of the material. The opposite trend is apparent at short timescales ($\tau/\tau_c(t_{age}) \ll 1$), and the strength of the relaxation spectrum increases with the relaxation mode $H(\tau, t_{age}) \sim G(t_{age})\,[\tau/\tau_c(t_{age})]^{\alpha}$, and hence longer modes and larger entities constitute higher shares to the relaxation process. As classified by Winter (14) and also discussed by Zaccone et al. (44), negative and positive power-law exponents in the relaxation spectrum represent gel- and glass-like viscoelastic behavior, respectively. Our results thus clearly demonstrate that these weakly aggregating gels display gel-like characteristics on long timescales but soft glassy behavior at short timescales.

This coexistence thus motivates the following picture: At sufficiently large length scales and long timescales the network of aggregated fractal clusters (initially formed at the gel point through RLCA) still shows a scale-free viscoelastic response similar to a critical gel, whereas at smaller scales the local jamming/arrest of small aggregating clusters leads to glassy behavior. A range of microstructural mechanisms, e.g., phase separation through spinodal decomposition followed by local coarsening (5) or jamming and double-ergodicity breaking (45), may be responsible for the onset of jamming/arrest of the material at smaller scales. The demarcation between these two regimes shifts to progressively longer times and larger length scales as the gel matures.

Close to the gel point (the right side of the blue shaded region in Fig. 4*C*) the underlying fractal network of aggregates shows a critical gel-like response. As the characteristic timescale $\tau_c(t_{age})$ grows, the local relaxation spectrum reaches a maximum $H(\tau = \tau_c, t_{age}) = H_{max} \sim [G(t_{age})/\pi]\tan(\pi\alpha/2)$ and the viscoelastic nature of the gel undergoes a transition to glassy behavior due to the local arrest in the gel structure that initiates from small clusters and progressively moves to larger scales (red shaded region in Fig. 4*C*). The characteristic timescale $\tau_c(t_{age})$ sets the extent of the glassy region and τ_c increases rapidly and monotonically with the system's age. As progressively larger entities become involved in the arrested dynamics, the aging in mature gels further slows down leading to ergodicity breaking. This translates into a progressively wider range of frequencies over which the glass-like behavior can be observed in the viscoelastic spectrum. Because τ_c increases linearly (or faster) with age time it becomes impossible to see back toward the initial gelation event, and ultimately the entire accessible relaxation spectrum resembles a soft glass with $G'' \sim (\omega\tau_c)^{-\alpha}$. However, fractal remnants of the critical gel persist in the structure (12, 13) and are reflected in the exponent α that is set at the gel point itself.

Conclusion

Coupling time-resolved mechanical and structural spectroscopy provides a unique perspective into the formation of weakly attractive colloidal gels. Using the concept of time–connectivity superposition we show how to generate a universal master curve and probe the viscoelastic response of these materials over a wide range of frequency and timescales spanning eight or more decades of magnitude. A simple three-parameter FMG framework describes this universal behavior and captures the symmetric power-law behavior observed in both the loss modulus and relaxation spectrum of the aging gel. The symmetric relaxation time spectrum $H(\tau, t_{age})$ (cf. Eq. **4**) reveals the dual nature of these weak colloidal gels: On large scales the decreasing power law ($\sim\tau^{-\alpha}$) is characteristic of gel-like behavior in the network, whereas on shorter timescales the relaxation spectrum shows an increasing power law ($\sim\tau^{\alpha}$), which is the hallmark of glassy behavior. The separating point τ_c between these two behaviors also increases with time as the gel stiffens and ages (cf. Fig. 3). However, despite the very significant temporal changes in the mechanical properties of these gels, the overall shape of the relaxation spectrum remains self-similar and is set by the value of the power-law exponent characterizing the relaxation dynamics at the gel point. For these weak colloidal gels, constituted from attractive particles with a small but non-negligible interparticle repulsion force, the fractal remnants of this connectivity persist long beyond gelation and the slow irreversible aging of the gel is captured through temporal evolution of the elastic modulus $G(t_{age})$ and the characteristic timescale $\tau_c(t_{age})$ of the material. Our findings shed light on the evasive and continuously evolving properties of soft glassy materials from early times beyond their formation and reveal an underlying dual viscoelastic nature that sets the rheological behavior of these aging materials over very wide ranges of time and frequency scales.

Materials and Methods

Sample Preparation. Aluminosilicate gels were prepared from two commercial solutions provided by Woellner Group GmbH: sodium silicate (Betol 39T) and sodium aluminate (Stabisil). The gels were prepared with a final concentrations of [Si] = 1.125 mol·L^{-1}, [Na] = 1.31 mol·L^{-1}, and [Al] = 0.12 mol·L^{-1} (referred to as the stoichiometrically balanced gel in the main text) and [Si] = 1.125 mol·L^{-1}, [Na] = 0.84 mol·L^{-1}, and [Al] = 0.12 mol·L^{-1} (referred to as the stoichiometrically unbalanced gel in the main text). More specific information on the material chemistry can be found in *SI Appendix*. To initiate the chemical reactions and gelation, both solutions with the desired chemical composition were rapidly mixed immediately before the experiments. The Ludox gel was prepared from silica nanoparticles (Ludox TM 50; Sigma-Aldrich) dispersed at a volume fraction of 3.7% in an aqueous solution containing urea (1 M) (41).

TR-SAXS. TR-SAXS measurements were performed on the SWING beamline at the SOLEIL Synchrotron (Saint Aubin, France). TR-SAXS spectra were recorded with a variable frequency during the gelation process. Initial spectra were measured every 100 ms to capture the fast initial kinetics and the acquisition rate was subsequently adapted according to the successive steps of the gelation process. The scattering was recorded with an Eiger 4M (Dectris) detector at a sample-to-detector distance of ~3 m, over a momentum transfer range of $0.0045 \leq q \leq 0.4$ Å^{-1} with X-ray energy of 12 keV and a wavelength of $\lambda = 1.03$ Å. The silicate and aluminate solutions were mixed in situ using a syringe pump and two isovolumetric syringes. The syringes, T-connection, and SAXS glass capillary were all connected with flexible polytetrafluoroethylene capillary tubes. The T-connection was used to provide homogeneous in situ mixing. The mixed sample was injected into a 2-mm glass capillary in the X-ray beamline for measurements. All spectra were normalized to the intensity of the transmitted beam, radially

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averaged, and background-subtracted using the local dedicated program Foxtrot (46). More specific information on the device and data analysis can be found in *SI Appendix*. USAXS/SAXS data for the mature gel were collected on an APS 9ID USAXS/SAXS/WAXS instrument (47, 48). The X-ray energy was 21 keV and data collection times were 90 s for USAXS and 30 s for SAXS. Data were fully background-corrected using the same measured container + solvent (water) scattering and placed on an absolute intensity scale using instrument-provided software (47–49). USAXS data were desmeared using established desmearing routines (50). USAXS + SAXS data were merged together yielding data over about four decades in q space $10^{-4} \leq q \leq 1$ Å^{-1}.

TR Rheology in the Linear Domain. Recently, inspired by advances in radar technology, we have employed exponential chirp signals for characterizing soft time-evolving (or "mutating") materials such as colloidal and polymeric gelling systems (22). While classical oscillatory rheometry imposes a monochromatic constant probe frequency at any point in time, in a chirp signal, the instantaneous frequency continuously varies, and hence chirp signals are also commonly referred to as sine sweeps. With only one carefully constructed signal a snapshot of the viscoelastic properties of a time-evolving material can be probed rapidly over a range of frequencies (*SI Appendix*, Fig. S12). For rheological characterization there is particular interest in exponential chirps, i.e., sine sweeps with constant amplitude but a probe frequency that increases exponentially in time. However, as Geri et al. (22) have recently shown, chirp signals can suffer from spectral leakage and other signal processing artifacts compromising the accuracy of the final measurements. To circumvent these errors they combined an exponential chirp signal with a smooth window function to significantly reduce the leakage error. They resulting protocol is known as an OWCh signal and results in a strain signal of the form

$$\gamma(t) = w(t)\gamma_0 \sin\left\{\frac{\omega_1 T}{\log(\omega_2/\omega_1)}\left[\exp\left(\log(\omega_2/\omega_1)\frac{t}{T}\right) - 1\right]\right\}, \qquad [5]$$

where T is the total length of the chirp signal and ω_1 and ω_2 are respectively the initial ($t = 0$) and final ($t = T$) frequencies attained by the signal. The window function $w(t)$ is a particular cosine-tapered waveform known as the Tukey window:

$$w(t) = \begin{cases} \cos^2\left[\frac{\pi}{r}\left(\frac{t}{T} - \frac{r}{2}\right)\right], & \frac{t}{T} \leq \frac{r}{2} \\ 1, & \frac{r}{2} < \frac{t}{T} < 1 - \frac{r}{2} \\ \cos^2\left[\frac{\pi}{r}\left(\frac{t}{T} - 1 + \frac{r}{2}\right)\right], & \frac{t}{T} \geq 1 - \frac{r}{2} \end{cases}, \qquad [6]$$

where r is a dimensionless tapering parameter that tunes how rapidly the amplitude is modulated within the length of the signal T. Geri et al. (22) show that for commercial rheometric hardware an optimum value of $r = 0.1$ minimizes the spectral leakage error and avoids excessive reduction of the power in the signal.

Fractional Derivatives and Spring-Pot Element. A spring-pot is a rheological element with the following constitutive relationship between stress σ and strain γ (28):

$$\sigma = \mathbb{V}\frac{d^\alpha \gamma}{dt^\alpha}. \qquad [7]$$

Here $\mathbb{V}$ is a so-called quasi-property that sets the strength of the gel and d^α/dt^α is the fractional derivative operator (based on the Caputo definition) with an exponent $0 \leq \alpha \leq 1$ (details in *SI Appendix*). In the limit $\alpha = 0$ we recover a purely elastic Hookean response (with $\mathbb{V} \to G$) and for $\alpha = 1$ a purely linear viscous material response (with $\mathbb{V} \to \mu$). For $0 < \alpha < 1$, this linear constitutive viscoelastic model predicts power-law frequency dependence in both moduli $G'(\omega) = \mathbb{V}\cos(\alpha\pi/2)\omega^\alpha$, $G''(\omega) = \mathbb{V}\sin(\alpha\pi/2)\omega^\alpha$, which results in a constant, frequency-independent phase angle $\delta = \alpha\pi/2$. Thus, this single-element fractional model, with only two fitting parameters $(\mathbb{V}, \alpha)$, is often considered an ideal representation for gelling systems at the critical gel point (10). As shown in Fig. 2*C*, for the unbalanced gel $\tan\delta \geq 1$ corresponding to $\alpha > 1/2$ before gelation (consistent with a liquid-like behavior), while $\tan\delta = 1$ at the gel point, which means that $\alpha = 0.5$ in our fractional constitutive model for this gel.

Data Availability. All study data are included in the article and/or *SI Appendix*.

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